

# FACET

Coordination analysis, cut by bond valence

A coordination number is not measured. It is produced by choosing a cutoff, and where a structure gives no gap for that cutoff to fall in, the number reported says as much about the person reporting it as about the crystal.

FACET cuts by bond valence rather than by distance, and reports not one coordination number but the range of threshold each one survives — so that a number which is a property of the structure can be told apart from one that is a property of the choice.

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# Getting started

FACET opens a crystal structure, works out what is bonded to what, and shows how that answer depends on where the line between bonded and not-bonded is drawn.

1. **Open a structure.** **File › Open structure**, or drag a file onto the window. CIF, mmCIF, VASP POSCAR/CONTCAR, XYZ, SHELX `.res` / `.ins`, PDB, VESTA and CrystalMaker files are read. **File › Open a folder of CIFs** loads many at once; the **Structures** dock on the left switches between them and can show several overlaid.
2. **Pick a site.** Click an atom in the 3D view, or choose a row in the **Sites** dock. The **Site** tab on the right then describes that site.
3. **Move the threshold.** The strip along the bottom is the cutoff explorer. Drag it and the coordination number, the bond-valence sum, the contact table and the drawn bonds all follow, with no recalculation of the neighbour search.

Nothing is loaded from the internet, and nothing is written anywhere until you export it.

### Worked examples

Six structures are bundled with FACET, under **File › Examples**. The six tutorials that follow use nothing else, so they can be worked through on a new installation with no files to find and no network. Each one states the numbers you should see, so you can tell whether you are reading the application correctly.

# Tutorial 1 · A coordination number you can trust

Quartz, $SiO_2$. Five minutes. Teaches the workspace, the Site tab and the cutoff explorer.

1. Open **File › Examples › Quartz ($SiO_2$)**. The structure appears in the 3D view and the **Structures** dock on the left gains a row.
2. In the **Sites** dock below it, click **Si**. The **Site** tab on the right now describes that site, and the silicon's polyhedron is drawn.
3. Read the top of the Site tab. You should see a **coordination number of 4** and a **bond-valence sum near 4.18** against a nominal +4.
4. Look at the contact list lower down: four oxygens, two at 1.605 Å and two at 1.611 Å, each carrying about 1.0 v.u., then a long gap before anything else.
5. Now the point of the application. Drag the vertical line in the **cutoff explorer** along the bottom of the window, left and right. Watch the coordination number in the Site tab.

**What you should see.** It stays at 4. The staircase for this site has one very long tread: the answer survives from 0.020 v.u. up to 1.035 v.u., a range of **1.71 decades** — a more than fiftyfold change in where the line is drawn. The Site tab reports that as the plateau width.

That is what it looks like when a coordination number is a property of the structure. Nobody arguing about quartz is arguing about whether silicon is four-coordinate, and the staircase shows why: there is no choice of threshold, within reason, that would give any other answer.

The sum of 4.18 rather than exactly 4.00 is the normal spread of tabulated $R_0$ values, not a fault in the structure. The Site tab shows the propagated uncertainty beside it.

## Tutorial 2 · And one you cannot

Eulytite, $Bi_4(SiO_4)_3$. Ten minutes. Teaches the contact list, the plateau, and the bond-valence vector. Tutorial 1 first.

The useful thing about this structure is that both kinds of site are in it, so nothing can be blamed on the data.

1. Open **File › Examples › Eulytite**.
2. Select **Si** in the Sites dock. Coordination number 4, four oxygens all at 1.623 Å, plateau **1.70 decades**. This is quartz again: a settled answer.
3. Now select **Bi** in the same structure. Coordination number **6**, bond-valence sum 3.24 against +3.
4. Read the contact list. Three oxygens at **2.158 Å** and three at **2.606 Å**. There is no third group, and nothing obvious separating the second group from the first.
5. Drag the cutoff explorer to the right, past 0.25 v.u. The coordination number drops from 6 to **3**, and three bonds disappear from the 3D view.
6. Turn on **View › Show bond-valence vector**. A lobe appears on the bismuth, pointing away from where the bonding is concentrated.

**What you should see.** The Bi plateau is 1.09 decades wide, running from 0.020 to 0.248 v.u. Above 0.248 the answer is 3; below it, 6. Both are defensible, and the published literature contains both. φ for this site is **0.419**, against 0.000 for the silicon in the same crystal: the bismuth environment is strongly one-sided, which is the $6s^2$ lone pair.

Compare the two sites' entries in the cutoff explorer. The silicon's staircase has one long tread; the bismuth's has several short ones. FACET will not choose between them for you, and that refusal is the point: reporting "six-coordinate" without saying at what threshold leaves out the only part of the statement that was a decision.

## Tutorial 3 · One compound, two answers

Senarmontite and valentinite, both $Sb_2O_3$. Ten minutes. Teaches loading several structures and exporting a table.

1. Open **File › Examples › Senarmontite**, then **File › Examples › Valentinite**. Both now sit in the **Structures** dock; click a row to make one active.
2. With senarmontite active, select the **Sb** site. Coordination number **6**: three oxygens at 1.978 Å and three at 2.902 Å. The plateau is **0.61 decades**, which is narrow.
3. Switch to valentinite and select its **Sb**. Coordination number **5**, at 1.977, 2.019, 2.022, 2.519 and 2.619 Å, with a plateau of **0.94 decades**.
4. Open the **Tools** tab on the right and look at **Cutoffs** and **Stability**, which tabulate this for every site at once.
5. **File › Export › Sites (CSV)** writes the active structure's table; repeat for the other and compare them in a spreadsheet.

**What you should see.** The same compound, determined by different people from different crystals, gives 6 in one polymorph and 5 in the other at the same threshold. Neither is an error. The bond-valence sums agree closely — 3.20 and 3.15 — because antimony collects the same total valence either way; what differs is how many neighbours it spreads it over.

This is the comparison to keep in mind when reading a coordination number out of a paper: it is a statement about one refinement under one rule, and the rule is usually not stated.

## Tutorial 4 · A number not safe to quote

Cryolite, $Na_3AlF_6$. Ten minutes. Teaches non-oxide work, the Stability table and the anion view.

Nothing in the method is specific to oxygen. This matters, because the 3.0 Å cutoff that floats around the oxide literature has no standing in a fluoride, and the bond criterion used here reaches a different distance for every pair.

1. Open **File › Examples › Cryolite**.
2. Select **Al**. Coordination number 6, sum 2.95 against +3, plateau **1.38 decades**. An octahedron nobody would dispute.
3. Select **Na1**. Coordination number 6, plateau 0.99 decades. Still comfortable.
4. Select **Na2**. Coordination number 6 — and a plateau of **0.10 decades**, running only from 0.0625 to 0.0779 v.u.
5. Drag the cutoff explorer slowly across that site. At 0.0625 the answer is **7**; by 0.080 it is **5**; by 0.090 it is **4**.
6. Open **Tools › Stability** to see all three sites ranked by how much threshold their answer survives, and **Tools › Anions** to look at the same structure from the fluorines instead.

**What you should see.** A factor of 1.4 in the threshold takes Na2 from seven-coordinate to four-coordinate. There is no plateau to stand on, so there is no coordination number to quote — only a coordination number **at a stated threshold**, which is a different and much weaker claim.

Al and Na1 in the same file have wide plateaus. The width is a property of each site, not of the structure as a whole, and the Stability table is where to look before quoting any of them.

## Tutorial 5 · One ion, two hosts

Eulytite and $BiPO_4$. Ten minutes. Teaches comparison across files and the **a priori** valences. Tutorial 2 first.

How much of what you measured belongs to the ion, and how much to the compound it is sitting in? Two files with the same Bi(III) in them answer it.

1. Open **File › Examples › Eulytite** and **File › Examples › Bismuth phosphate**.
2. In eulytite, select **Bi**: coordination number 6, φ **0.419**, bonds from 2.158 to 2.606 Å.
3. Switch to $BiPO_4$ and select **Bi**: coordination number **8**, φ **0.128**, bonds from 2.311 to 2.753 Å. The sum is much the same, 3.04 against 3.24.
4. Turn on **View › Show bond-valence vector** and switch between the two structures. The lobe is long in eulytite and short in the phosphate.
5. Open **Tools › A priori** in each. This solves the network equations from the connectivity alone, before any bond length is used, and reports how much of the asymmetry the topology alone requires.

**What you should see.** The same ion, at the same oxidation state, collecting the same total valence, but over six neighbours in one host and eight in the other, and far more one-sidedly in the silicate. Phosphorus competes harder for the oxygens than silicon does, and leaves bismuth a more even, more distant shell.

So "the coordination number of Bi(III)" is not a property of bismuth. It is a property of bismuth in a named compound, under a stated rule.

## Tutorial 6 · Making a figure for a paper or a talk

Ten minutes. Teaches themes, orientation and vector export.

1. Load any structure and select a site worth showing.
2. Choose a theme from **View › Theme**. **Light (for print)** and **Publication (greyscale)** are built for paper; **VESTA-like (white)** will look familiar to most readers. **More colour settings…** opens the Theme tab, where individual elements can be recoloured and the result saved and shared with a group.
3. Orient the structure. **View › Along a**, **b** or **c** gives an exact axis view, which is reproducible in a way that dragging is not.
4. Set the drawing style and polyhedra from the two rows of controls above the 3D view, and choose how many cells to tile.
5. **File › Export vector…** writes **SVG** or **PDF** with real geometry — lines and polygons that stay sharp at any magnification and can be edited in Illustrator or Inkscape. Use **Export image…** only when a bitmap is specifically wanted.

**A note on the vector export.** It is produced by the software renderer, which draws the same scene with the same geometry as the hardware one. A figure exported on a machine with no graphics card is identical to one exported on a workstation, which is the reason that renderer exists.

For a table rather than a picture, **File › Export** also writes the sites and contacts as CSV or XLSX, the cutoff table, and a threshold scan — the last being the data behind the staircase, if you would rather plot it yourself.

### A report on many files

**File › Export › Report on CIFs…** takes a set of files or whole folders, analyses every site of every file at the thresholds you give (the window's own are offered), and writes five tables: **sites** (coordination number, the bond-valence sum over the bonded set and to the listing cut, phi, the polyhedron's distance statistics, the plateau the threshold sits on), **contacts** (every distance with its valence, occupancy, running sum and whether it counts as a bond), **plateaus** (the whole staircase of each site as rows), **cutoffs** (the distance each cation–anion pair's thresholds correspond to) and **balance** (the cation and anion totals of each structure). The same tables go out as an Excel workbook, CSV files (one per table), JSON, a web page, Markdown or plain text, each headed by the thresholds, the parameter set, the version, the time and the list of files. A file that cannot be read stays in the report with the reason, so a short table is never silently short.

The same report from the command line, with no window:

```
py -3.11 -m facet.report structures/ -o report.xlsx
py -3.11 -m facet.report a.cif b.cif --format html --v-bond 0.10 -o cmp
```

Two more commands run without the window. `py -3.11 -m facet.site FILE` prints one site's analysis (`--all` for every site, `--format md` or `tex`). `py -3.11 -m facet.check` recomputes every number the tutorials state for the bundled examples on this machine and reports each against its tolerance; `--parity TABLE TRAJECTORY` lays another code's per-atom values for one MD frame beside FACET's, descriptor by descriptor.

## The bundled structures

Six structures ship inside the application, under **File › Examples**. They are there so the tutorials can be followed on a new installation with nothing downloaded, and they are chosen to disagree with one another.

| | |
|---|---|
| **Quartz**<br>SiO2 · COD 9013321 | Silicon sits in a tetrahedron of four oxygens at 1.605 to 1.611 angstrom, and the next contact is far enough away that the answer four survives a threshold moved over more than a hundredfold range. This is what a plateau looks like when a structure has nothing to hide, and it is the comparison every other example here is read against. |
| **Eulytite**<br>Bi4(SiO4)3 · COD 9012894 | The silicon is as clean as quartz. The bismuth in the same crystal has three contacts near 2.16 angstrom and three near 2.61, with no gap between the two groups that a cutoff could be placed in without choosing it. Both sites are in the same file and the same refinement, so the difference is not a question of data quality. |
| **Senarmontite**<br>Sb2O3 · COD 9009747 | Antimony here is six-coordinate, three short bonds at 1.978 angstrom and three long at 2.902. Open it beside valentinite, which is the same compound, and compare the coordination numbers the two give at the same threshold. |
| **Valentinite**<br>Sb2O3 · COD 9007587 | The same Sb2O3, in a different arrangement. The antimony is five-coordinate and its plateau is wider than senarmontite's. Two polymorphs of one compound need not report the same coordination number, and neither answer is wrong. |
| **Cryolite**<br>Na3AlF6 · COD 9004097 | Nothing in the method is specific to oxygen, and this is where that matters: a 3.0 angstrom cutoff chosen for an oxide has no standing in a fluoride. Two of the three sites behave; the Na2 site has a plateau around a tenth of a decade wide, which means its coordination number is a property of the threshold rather than of the structure. FACET says so rather than picking one. |
| **Bismuth phosphate**<br>BiPO4 · COD 9008088 | The Bi(III) of eulytite again, with phosphorus competing for the same oxygens instead of silicon. The coordination number rises, the bonds lengthen, and the bond-valence vector shortens considerably. Useful for asking what belongs to the ion and what belongs to the compound. |

All six come from the Crystallography Open Database, whose contents are in the public domain, and are redistributed unchanged. The original determination for each is credited in `ATTRIBUTION.md` beside the files; cite the paper rather than the database. Nothing stops you passing FACET on with them included.

**Open all of them** at the foot of the menu loads the whole set at once, after which the **Structures** dock switches between them.

## Why valence, not distance

A coordination number is not measured; it is produced by a rule. The commonest rule is a distance cutoff — "everything within 3.0 Å" — and the number it gives changes when the cutoff does. Two papers on the same compound can report different coordination numbers without either having made an arithmetic error.

FACET therefore cuts on **partial bond valence** instead:

$$v = \exp((R_0 - d) / b)$$

with $b$ = 0.37 Å and $R_0$ a tabulated constant for the element pair and oxidation states. Inverting it,

$$d = R_0 - b \ln v$$

so one valence threshold gives a **different** distance for every pair. At the shipped 0.075 v.u. a contact counts as a bond out to $R_0$ + 0.958 Å; at the 0.02 v.u. tabulation threshold it is listed out to $R_0$ + 1.447 Å. A short Si–O bond and a long Bi–I bond are then being judged by the same physical criterion rather than by the same number of ångströms. The **Cutoffs** tab under **Tools** prints the distance each pair gets, so the translation is visible rather than implied.

### What FACET does not do with this

It does not tell you which threshold to use, and it does not mark a coordination number as the right one. What it reports instead is the **plateau**: the range of threshold over which a given count holds. A count that survives a wide range of threshold and one that changes if the threshold moves in the fourth decimal place are different situations, and the width is the statement of which you have.

The uncertainty on $R_0$ is systematic: it scales a whole bond-valence sum rather than averaging out. FACET propagates it and prints it beside the sum.

## The workspace

### Left

**Structures** — every file loaded in this session. Tick to show, use the overlay toggle to draw several in one view, and the cross to remove one.
**Sites** — the crystallographic sites of the active structure. Selecting one drives the Site tab, the polyhedron and the cutoff explorer.

### Centre

The 3D view, with two rows of controls above it: drawing style, polyhedra, how many cells to tile, the unit cell, the projection, what to label atoms and bonds with, and the tabulation threshold.

### Right

Ten tabs. **Site** is the analysis of the selected site. **Tools** holds the whole-structure tables. **Diffraction** computes a powder pattern, **PDF** a pair distribution function, and **EXAFS** the shell list and resolution report. **Planes** cuts lattice planes and slabs. **Styles** lists per-atom and per-site styling. **Volume** maps bond-valence sum through space. **Disorder** handles partial occupancy and alternative configurations. **Theme** is colour, sizing and palettes.

Most controls state their own name inside themselves rather than beside them -- a spin box reading "r to 20.0 Å" is the r limit. Hovering any of them explains what it does, and the settings that are set once and left sit behind a disclosure arrow.

### Bottom

The **cutoff explorer**. Each site is a staircase: coordination number against threshold, with the flat treads being the plateaux. Drag the vertical line to move the bond threshold; everything above follows it — the drawn bonds, the labels on the atoms, the site list, the Site tab, the tables under Tools and the report. The contacts were found once, down to the tabulation threshold; moving the cut only decides which of them count as bonds, so a drag re-sums them and searches for nothing.

A round-headed **pin** marks the threshold the file was opened at, so after a drag the numbers first shown for it can be found again. The pin is where the structure was first analysed, not a recommendation.

The **right axis** carries the running bond-valence sum over the contacts counted so far, in green, with the size of the site's charge drawn across it for scale. The area under the sum is shaded, and so is the step the next contact outside the cut would add, with the increment written on it. The staircase and the sum step at the same contacts: read them together to see what each successive neighbour adds, and where the next one would take the sum; the line under the plot states that too. The corner button switches the horizontal axis between bond valence and distance; the two orders differ when a site has more than one kind of anion, and each axis shows its own.

As the window opens, the axis is distance, the staircase and the sum sit in stacked panes, the area under the sum is filled, one bar per contact runs under the rail, and the Site tab's contact table carries a running-sum column. **View › Cutoff explorer** holds every way of reading the same contacts, none of which changes a number: the axis choice; the fill; the sum as a **percentage of the site's charge**, so that sites of different charge share one scale; **stacked panes**, the staircase above and the sum below on their own axes; **one bar per contact** under the rail, each contact's own contribution with the running sum above it; a **running-sum column** in the Site tab's contact table; and a shortcut to the bond-label mode that writes on each drawn bond its share of the sum.

**Thin sticks for the contacts under the threshold**, in the same menu, is on by default: the contacts between the listing threshold and the bond threshold are drawn as thin faded sticks, with no share of the sum and no label, so the picture shows what the cut left out as well as what it counted. Off draws only the bonds the threshold counts; a drag of the cursor then changes the set of sticks once it settles. The choice is remembered.

**Focus the 3D view within the cut**, in the same menu, dims every atom but the selected site and those within the distance the threshold corresponds to for it — the explorer's distance axis read at the cursor — and draws that sphere, translucent, about the site. Drag the cursor and the sphere grows or shrinks with it; the atoms that come inside are drawn in full, with the bonds among them, so the coordination the cut gives is seen on the structure as well as counted. The status bar states how many atoms are inside and at what radius. The focused atom is the one you clicked when it belongs to the selected site, else the site's atom nearest the middle of the drawn block.

Both splitters can be dragged, and the docks can be undocked or closed. The window remembers nothing between runs, so a layout you liked has to be set up again — that is a limitation, not a preference.

### An MD model

When the row selected in the Structures dock is an MD model, the same five regions show the Model workspace instead: an Elements dock, the frame stepper, the frame or a figure, the threshold strip, and the Setup, Results, Highlight and Notes tabs. See **MD models and the Model workspace**.

## Reading the Site tab

Everything on this tab is a measurement of the selected site at the current threshold.

| | |
|---|---|
| **Coordination number** | Contacts whose partial valence exceeds the bond threshold. |
| **Bond-valence sum** | Σ**v** over those contacts, occupancy-weighted, with the propagated $R_0$ uncertainty and the difference from the site's nominal oxidation state. |
| **ECoN (Hoppe)** | Effective coordination number: a weighted count that needs no cutoff at all, shown for comparison with the thresholded one. |
| **Maximum-gap split** | The count you get by cutting at the largest relative gap in the sorted distance list. Where no gap is larger than the tolerance, the full contact list is reported rather than an arbitrary split. |
| **φ** | $\lvert\Sigma \mathbf{v}_i \hat{\mathbf{u}}_i\rvert / \Sigma v_i$ — the length of the bond-valence vector sum divided by the scalar sum. Zero for a centrosymmetric environment, and larger when the bonding is one-sided. It is dimensionless and exactly invariant to an error in $R_0$, because a common factor cancels between numerator and denominator. |
| **Void cone half-angle** | The widest cone about the site that contains no bonded ligand direction. |
| **Plateau at this threshold** | How far the threshold can move without the coordination number changing, in decades of valence and in ångströms of distance. |
| **Microscopic optical basicity λ** | For an anion site: Duffy and Ingram's λ from the bond valences it receives, with each cation's share listed below the notes. For a cation site: the mean λ of the anions about it. See **Optical basicity**. |

Below that, every contact found at the tabulation threshold, with its distance, its partial valence, a filled marker if it is above the bond threshold and an **est.** flag if its $R_0$ was estimated rather than fitted. The footer states the $R_0$, **b** and source actually used.

# The bond-valence vector, and the lone pair

**View › Show bond-valence vector** draws a lobe on each cation pointing away from its bonded ligands.

### What is drawn

For each drawn atom, FACET forms $\mathbf{V} = \Sigma v_i \hat{\mathbf{u}}_i$ over the contacts above the bond threshold, where $\hat{\mathbf{u}}_i$ points from the cation to the ligand. The lobe is drawn along $-\mathbf{V}/|\mathbf{V}|$ — the direction the bonding is **not** — and its length is φ × the site's mean bond length × a display scale you set. So a centrosymmetric site draws nothing at all, because φ is zero there.

### What that is and is not

It is the direction and relative size of the bond-valence vector sum, which is a measurement of the coordination geometry. For a cation with an **ns**$^2$ configuration this direction is where an ns$^2$ lone pair is conventionally described as pointing, which is why the feature is asked for — but the arrow is the vector sum, not an electron density, and its length is a drawing scale rather than a distance to anything. FACET does not place a pseudo-ligand at a literature lone-pair distance: that number would come from elsewhere while looking like a measurement of your file.

The cone overlay is the other half of the same picture: a real cone of the measured void half-angle about the void axis. Where several equally wide cones exist, one axis is returned and the degeneracy is stated in the note.

Both follow the bond threshold, so a lobe that changes markedly as you drag the explorer is telling you that the vector sum depends on contacts near the cut.

# Tools

Whole-structure tables, each recomputed for the active structure.

- **Cell** — lattice parameters, the reciprocal cell in the crystallographic convention ($\mathbf{a}^* = \mathbf{b}\times\mathbf{c}/V$, without $2\pi$), density, composition and formula weight.
- **File** — where the structure came from, what every health check measured, and the displacement parameters. See below.
- **Angles** — ligand–centre–ligand angles for the selected site, over the bonded set.
- **Shells** — neighbours grouped into shells with their degeneracies, which is the form a radial analysis wants.
- **Reflections** — $d$-spacings and Bragg angles for a chosen wavelength. Geometry only; intensities live in the Diffraction tab.
- **CN methods** — the same site counted by every rule FACET can apply: valence-thresholded, ECoN, maximum gap, Voronoi, and optionally eight pymatgen strategies. They are listed side by side, and none is marked as the answer.
- **Connectivity** — which polyhedra share one ligand, two, or three and more: corner, edge and face sharing.
- **Cutoffs** — the distance each element pair gets from the current thresholds, with its $R_0$, $b$, and where they came from.
- **Stability** — the plateau for every site at once: from where to where the threshold can move before that site's count changes.
- **Anion valences** — bond-valence sums computed **around each anion**, by a search in the opposite direction from the cation one. Cation and anion totals count the same bonds from opposite ends, so their agreement is a check on the arithmetic rather than a restatement of it. The charge balance line reports the residual.
- **Basicity** — the optical basicity: $\Lambda$ from the composition, the mean microscopic $\lambda$ of the anions from their bond valences, every anion and cation site as rows, and the moderating parameters on the chosen scale. See **Optical basicity**.

A tab is computed when you show it, not before. Recomputing all ten on every move of the threshold took nearly four seconds on a structure of a few hundred atoms, and nine of them were behind another tab at the time.

### The File tab

What the file says about itself, and what can be measured about it. Three parts:

**Where this came from** — the path, the name, the formula as written, the space group, the database code and the reference. Every number on every other tab is derived from this file, and the file is not always the one you think it is.

**Checks** — coordinates that collide, occupancies over one, uncertainties large enough to swallow a bond length, a formula that disagrees with the atom list, a bond-valence sum far from the stated oxidation state, a displacement tensor that describes no ellipsoid. Each says what it measured and the value it was compared against, grouped by how far outside the ordinary range it fell. **None of them says whether the structure is usable.** That depends on what is being asked of it: coordinates too uncertain to settle a bond-valence sum still carry the cell that indexes a powder pattern.

**Displacement** — where the file gives anisotropic parameters, the tensor converted to Cartesian axes and diagonalised. $U_{eq}$ is one third of the trace, which is the quantity comparable with a quoted $U_{iso}$; the three r.m.s.

values are the displacements along the principal axes. That last part is what no single number carries — a site can have an entirely ordinary $U_{eq}$ and still be four times longer than it is wide.

A displacement tensor is a covariance matrix and has to be positive definite. One that is not describes a hyperboloid, which is not a shape an atom can have, and the site was refined where the data did not constrain it. Such a site is listed with its eigenvalues rather than with an r.m.s. it does not have. The test is made on the file's own components, so it does not depend on the conversion: changing basis cannot change the sign of an eigenvalue.

# Diffraction and the pair distribution function

## Powder pattern

The **Diffraction** tab computes $|F|^2$ from the structure for X-rays, neutrons or electrons, applies the Lorentz–polarisation factor and multiplicity, and convolves with a pseudo-Voigt whose width follows the Caglioti form $\mathrm{FWHM}^2 = U\tan^2\theta + V\tan\theta + W$. A measured pattern can be loaded and scaled to the calculation by least squares, with the difference curve underneath.

Exact from the structure: the reflection positions, the multiplicities, the structure factors given the tabulated form factors. Assumed: the wavelength, the peak shape and widths, and the isotropic displacement parameter used where the file gives none. The notes under the plot say which of those applied.

## Pair distribution function

The **PDF** tab computes $G(r)$ from the same structure:

$$R(r) = (1/N)\,\Sigma_{i\neq j}\, w_{ij}\; N(r;\, r_{ij},\, \sigma_{ij}) \quad , \quad G(r) = R(r)/r - 4\pi r\rho_0$$

with $w_{ij}$ the scattering weights for the radiation chosen. The area under a peak of $R(r)$ is the scattering-weighted coordination number for that shell, which is the quantity a PDF is usually read for — so it is printed for the peak you select.

Two instrument effects are offered, and both are off until you set them. Truncating the transform at a finite $Q_{max}$ is a convolution in $r$, and it is what produces the ripple below the first peak; FACET applies it over the odd extension $G(-r) = -G(r)$, so that ripple appears where a measurement would show it. Finite $Q$-resolution is the dual operation, a multiplication by $\exp(-(rQ_{damp})^2/2)$.

The peak widths come from the displacement parameters in the file: $\sigma_{ij}^2 = U_i + U_j$ for isotropic $U$. That is the **uncorrelated** width. Near neighbours move together, which makes their peaks narrower than this, and the correlation terms ($\delta_1$, $\delta_2$) are user inputs defaulting to zero — FACET cannot measure them from one structure.

This is the PDF of the average crystal. A fractionally occupied site contributes fractional pair correlations, and a real disordered material's measured PDF differs from it by design — that difference is usually the reason a PDF was measured.

# EXAFS shells and FEFF

The **EXAFS** tab turns the structure into the things an EXAFS fit needs, and reads back what a real scattering calculation produced.

### The shell table

Per shell about the absorber: element, degeneracy **N** (and the occupancy-weighted **N**), mean **R** with the spread inside the shell, which crystallographic sites contribute, and two independently sourced estimates of $\sigma^2$ — the uncorrelated value from the file's own displacement parameters, and an Einstein-model value for a correlated pair at a temperature and Einstein frequency you set. They are different quantities and are labelled as such.

### The resolution report

Over the **k**-range you intend to fit, the resolution is $\Delta R = \pi/(2\Delta k)$, and the number of independent points is $N_{idp} = 2\Delta k \Delta R/\pi$. FACET states which of your shells are closer together than $\Delta R$, and how many parameters your shell list would need against $N_{idp}$. Both are statements about the structure and the range you chose.

### FEFF

**File › Export › FEFF input for this site** writes a `feff.inp` for the selected absorber. FACET does not run FEFF and does not bundle it; if you have a `feff8l` or `feff6l` executable, point the panel at it and the panel will run it and read the results back. `files.dat` and the `feffNNNN.dat` path files are parsed to show FEFF's own degeneracy, effective path length, path type and amplitude ratio beside FACET's crystallographic shells — which is how a single-scattering count and a real multiple-scattering calculation can be compared without either being asserted from the other.

### What is declined, and why

FACET does not compute χ(**k**) from the structure alone. A single-scattering χ(**k**) without phase shifts puts its Fourier transform peaks about 0.4 Å away from the distances that generated it, so a curve produced that way misstates the one quantity it would be read for. Where FEFF path files are present, χ(**k**) **is** built — from FEFF's amplitudes and phases with FACET's degeneracies and distances — and it is labelled with where each half came from. XANES needs full multiple scattering or a DFT calculation; FACET writes the input for one and does not pretend to substitute for it.

# Planes, slabs and volumes

## Planes

Enter (**hkl**) or pick from the presets to draw the lattice plane and report its **d**-spacing, the atoms lying on it within a tolerance you set, and its occupancy. Interplanar angles between two sets of indices are computed from the reciprocal metric, so they are right in a triclinic cell as well as a cubic one.

## Slabs

A slab keeps only what lies between two parallel planes. Bonds are kept only if both of their drawn endpoints survive, so a slab never leaves a bond hanging into empty space.

## Volume

The **Volume** tab evaluates the bond-valence sum a probe ion of your choice would have on a grid through the cell, and draws either a section through it — with contours and a colour map — or an isosurface in the 3D view. The section is where a migration path or an interstitial site shows itself as a region of near-correct valence for the probe.

Grid work is bounded: a request large enough to take minutes is refused with the number it would have needed, rather than freezing the window.

# Disorder and overrides

### Disorder

Where a CIF declares assemblies and groups, the **Disorder** tab lists them and lets one configuration be shown at a time. Where it declares only partial occupancies, sites closer together than a physical bond are reported as mutually exclusive candidates. Which configuration is real is not decided here.

### Styles

Any element, site or single atom can be given its own colour, radius or visibility. The **Styles** tab lists every one of them so that a figure can be retraced, and any of them can be removed. Overrides belong to the structure they were made on, and an atom index that no longer addresses an atom is dropped when a file is reloaded rather than applied to whatever landed on that index.

Everything on both tabs is undoable: **Edit › Undo**, or Ctrl+Z.

# Themes and figures

**View › Theme** switches the whole application — viewport, menus, panels and tables. The presets are a white VESTA-like default, a light theme for print, a greyscale publication theme with depth cueing off, a high-contrast theme chosen to survive the commonest colour-vision deficiencies, and two darker grounds for working at night.

The **Theme** tab goes further: per-element colours, what an atom's colour **means** (element, site, bond-valence sum, coordination number, φ, or valence discrepancy), atom and bond scaling, depth cueing, ambient occlusion and outlines. A theme can be saved to a file and shared, which is how a set of figures from several people ends up matching.

## Exporting a figure

**File › Export image** writes a raster image at a chosen supersampling. **File › Export as vector** writes SVG or PDF — real vector geometry, not a bitmap in a wrapper, so it can be edited in a drawing program and scales without pixels. The vector exporter is the same code path as the software renderer, so what it writes is what the fallback tier draws.

## Anion sites

By default the site list holds the cations, and the anions appear only as their ligands. **Anion sites**, under the list, analyses each anion as a site in its own right: its coordination number, its contacts, its plateau and its own bond-valence sum, counted from the cations around it.

It is the other half of the same check, and it catches things the cation sums do not. An oxygen whose sum comes to 1.4 is either missing a bond to something the file leaves out — a hydrogen, usually — or is not where the refinement put it. Cation and anion totals count the same bonds from opposite ends, so their agreement is a check on the arithmetic rather than a restatement of it.

The search is done around the anion rather than gathered from the cation results. A cation site's contact list belongs to one representative atom of that site, and an anion is reached by cations from every equivalent position — most of which are not that representative. Summing the cation lists gives each anion only the fraction of its bonds that happen to touch a representative, which on α-$Bi_2O_3$ came to 0.66 v.u. for an $O^{2-}$: low enough to look like a finding rather than an error.

## What the view turns about

By default the view rotates about the middle of the drawn cell block. Right- click an atom and choose **Rotate about <label>** to turn about that atom instead — useful when the site you are examining is at the edge of the cell and keeps swinging out of frame. The picture does not jump when you choose it: the camera pans so that the atom is where the middle of the view already was, and nothing grows or shrinks.

A site you cannot find in the view can be reached from the site list on the left: double-click it, or right-click it for the same two choices. Where a site is drawn several times over — once per cell of the block — the copy nearest the middle of the block is the one used, since turning about one at the edge puts the rest of the structure off to one side.

The toolbar says which atom it is for as long as it is set. **C** puts the centre back at the middle of the cell, **Esc** clears it along with the selection and any measurement, and panning with the right or middle button moves the centre off the atom as a pan always has.

The centre is kept as a position, not as an atom. Nothing that redraws the structure — moving the threshold, changing the style, cutting a slab — disturbs it, but if the same atom moves, as it can between disorder configurations, the centre stays where it was rather than following.

### Displacement ellipsoids

The **Style** list has an entry for them. Each atom is drawn as the surface enclosing a chosen fraction of its displacement distribution — 50% by default, which is the crystallographic convention and what every other program draws unless told otherwise. The fraction is chosen beside the style, and it is shown there rather than hidden in a menu because it is part of what the picture says: a 50% and a 99% ellipsoid of the same atom differ by more than a factor of two.

The lines across each ellipsoid are the boundaries of its octants, drawn where its three principal planes cut the surface. They are what makes it read as a solid with an orientation rather than as a flat oval, and they are the convention ORTEP set. An atom drawn without them is one the file gave no anisotropic tensor for: it is a sphere from $U_{iso}$, or, where there was nothing at all, a small fixed sphere. Those have no principal axes, and marking axes on them would assert a direction the file never gave.

Unlike every other style, the drawn size here is a measurement rather than a convention. It is not scaled by occupancy and not scaled by the theme's atom size, both of which would make it a picture of something else. Bonds are drawn thin in this style, because a 50% ellipsoid is about a tenth of an angstrom across and an ordinary bond is drawn wider than that.

A tensor that is not positive definite describes no ellipsoid at all. Such a site falls back to its isotropic parameter and is listed on the **File** tab under the checks, with the eigenvalue that makes it impossible.

### Lighting

The light is fixed to the viewer, not to the crystal: turning the structure does not turn the lamp with it, so looking along −c is lit exactly as looking along +c. Depth cueing, which fades distant atoms towards the background, is the separate control in the Theme tab, and it is what the publication theme switches off.

### Stereo

Anaglyph, greyscale anaglyph, side-by-side and cross-eyed pairs are in the Theme tab. The separation is an angle rather than a distance, so the depth survives zooming.

## Keyboard and mouse

| | |
|---|---|
| **Left-drag** | Rotate |
| **Middle-drag** or **right-drag** | Pan. This moves the centre of rotation with it. |
| **Wheel** | Zoom |
| **Left-click an atom** | Select that site |
| **Left-click a bond** | Its two atoms, length and valence, the share of the cation's sum it carries, the $R_0$ and **b** it was computed with and their source, and the pair's cutoff distance at the threshold in force: in a tip at the cursor, the status bar and the Messages dock |
| **Right-click** | Context menu for the atom under the cursor: colour, radius, visibility, its polyhedron, what to turn the view about; **Isolate …** alone, with its bonded neighbours, or with their neighbours too (the rest left out until **Show the whole structure again**); **Show the angles at …** draws the bond angles at it as arcs with their values — by default only between the bonds whose far atom lies nearer the viewer than this one, so a seven-fold site reads as a few arcs; turn the view to bring other bonds to the front, or untick **Angles of the front bonds only** for every pair; **Show the lone-pair lobe at …** draws this site's bond-valence vector on its own, with its φ stated (the site has to have one: φ above 0) |
| **on…** (toolbar) | Which atoms carry the labels chosen under **Label atoms** and **bonds**: every atom, or the sites and elements ticked, as many as you like; a bond is labelled when either end is ticked |
| **Click two atoms** | Distance |
| **Click three** | Angle |
| **Click four** | Torsion |
| **R** | Reset the view |
| **C** | Turn about the centre of the cell again |
| **Esc** | Clear the selection, the measurement and the rotation centre |
| **L** | Cycle the atom labels |
| **Ctrl+O** | Open a structure |
| **Ctrl+S** | Export an image |
| **Ctrl+Z** / **Ctrl+Y** | Undo / redo |
| **F1** | This manual |
| **Ctrl+Q** | Quit |

A drag that moves more than a few pixels is a rotation, not a click, so turning the structure and releasing over an atom does not select it.

Files can be dropped onto the window, one or many. An MD model or trajectory among them is listed in the Structures dock and opens the Model workspace (see **MD models**), where **Page Down** / **Page Up**, **Home** and **End** step through the frames and **H** opens the Highlight tab.

## What FACET does not do

Stated plainly, because a tool that is quiet about its limits gets used beyond them.

- **It does not refine anything.** No least-squares against measured data, no structure solution. Measured patterns can be overlaid and scaled, not fitted.
- **It does not judge a structure.** No number is labelled good, poor or trustworthy, and no coordination number is marked as the right one. Where a quantity depends on a choice, the choice is shown next to it.
- **It uses anisotropic displacement parameters only as isotropic equivalents** in the derived quantities, so PDF peak widths and $\sigma^2$ estimates use $U_{iso}$. The tensor itself is read and reported on the File tab, and is not yet drawn as an ellipsoid.
- **It does not compute XANES**, and it does not compute $\chi(k)$ from geometry alone. See the EXAFS section for what it does instead.
- **It ships one bond-valence table, not a compilation.** Brese & O'Keeffe's Table 2 — 108 cations against O, F and Cl — with a few pairs to the heavier anions from their Table 3. Larger accumulated sets, and the newer refits, are read from files you supply. A pair outside the table falls back to the O'Keeffe–Brese estimator and is labelled as estimated wherever it is used: the estimator differs from a fitted value by 0.05 Å on average and by as much as 0.21 Å, which is a factor of 0.57 to 1.48 on every bond valence.
- **It does not correct measured data.** No absorption, no background subtraction, no $Q$-space corrections.
- **It does not run molecular dynamics.** It reads the models and trajectories a simulation wrote and measures them; it integrates no equation of motion, fits no potential and edits no model. Nor does it simulate a quadrupolar NMR spectrum: its NMR analysis gives isotropic shifts from a published structure-shift correlation whose coefficients and reference are supplied by the user, none shipping with FACET.

### Where the numbers were checked

The repository carries `VERIFICATION.md`, which records how each quantity was checked against something with no code in common with it: an independent implementation, a closed form, a physical law, an invariance, or the input file's own statement of itself. Five faults invisible from inside the program were found that way, including a symmetry expansion that silently lost three quarters of one structure's oxygen while every internal check still agreed.

If a number here disagrees with one from elsewhere, the threshold and the parameter set are the first two things to compare — both are printed in every export.

# Licences and notices

FACET's own source code is under the **MIT Licence**, which places no restriction on passing it on.

The **built application** bundles third-party components, and one of them constrains how a binary may be distributed.

### Qt, under the LGPL-3.0

Qt is used under the LGPL-3.0. That is permitted for an application under any licence, but LGPL-3.0 §4 requires that whoever receives the binary be able to replace the Qt libraries with their own build and re-run it. FACET is therefore packaged as a **directory**, not a single self-extracting executable, so the Qt DLLs sit beside the program as ordinary replaceable files. A `licenses` folder next to the executable carries the LGPL text and a note on how to do the replacement.

### Everything else

| | |
|---|---|
| gemmi | MPL-2.0 |
| spglib | BSD-3-Clause |
| NumPy, SciPy, pandas | BSD-3-Clause |
| matplotlib | matplotlib licence (PSF-derived) |
| pymatgen | MIT |
| PyInstaller | GPL-2.0-or-later, with the bootloader exception that permits distributing a bundled application under any licence |
| Mesa llvmpipe ( `opengl32sw.dll` ) | MIT |

### Bond-valence parameters

The IUCr `bvparm2020.cif` table (I. D. Brown, McMaster University) is bundled unchanged under its own notice: it may be used and distributed without fee for non-profit purposes provided the notice is kept, no fee is charged for the table, and any change is recorded in the file. FACET makes no change to it. The individually cited rows it draws on (Brown & Altermatt 1985, Brese & O'Keeffe 1991, Gagné & Hawthorne 2015 and the rest) are literature values. The softBV set carries terms of its own and is not bundled; it is read from a file. FACET records which set produced a given number. Cite the set you used.

The full text, with the pre-distribution checklist, is `THIRD_PARTY_NOTICES.md` in the repository and beside the executable.